Mikhail Kornev

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RESEARCH AREAS

Algebraic Topology, Homotopy Theory, Group Theory, and Combinatorics

APPOINTMENTS

Sep. 2022 –	PhD student and research assistant at Steklov Mathematical Institute of RAS Department of Geometry and Topology
Mar. 2026 –	Research assistant at HSE University International Laboratory for Mirror Symmetry and Automorphic Forms
Sep. 2021 – 2025	Research assistant at HSE University Laboratory of Algebraic Topology and Applications

EDUCATION

Sep. 2016 – May 2022	Specialist degree Diploma with honors in Mathematics (equivalent to M.S.) Lomonosov Moscow State University

RESEARCH PAPERS

Mathematics

Apr. 2026	Kornev M.I., *Buchstaber, Ochanine, Krichever, and Witten genera* Journal of Geometry and Physics, Volume 226, Number 105855 (2026) https://doi.org/10.1016/j.geomphys.2026.105855
Jan. 2026	Buchstaber V.M., Kornev M.I. *n-Valued Monoids on $\mathbb{C}P^1$ and Discriminants* Russian Mathematical Surveys, Volume 81, Issue 1(487) (2026), 191–193 https://doi.org/10.4213/rm10280e
Jan. 2026	Kornev M.I. *On the Classification of n-Valued Monoids and Groups of Order* 3 Mat. Zametki, Volume 119, Issue 2 (2026), 244–252 https://doi.org/10.4213/mzm14803
Jan. 2024	Kornev M.I. *n-Valued Coset Groups and Dynamics* Mathematical Notes, Volume 116, Issue 1 (2024), 66–76 https://doi.org/10.4213/mzm14243
Sep. 2023	Kornev M.I. *Homology Spheres*

Transactions of the Moscow Mathematical Society, Volume 84 (2023), 197–239
https://doi.org/10.1090/mosc/352

Mathematics and Music

Sep. 2025 *The Technique of Musical Icosahedra*
Music Academy, no. 3, 2025, pp. 64–83
https://doi.org/10.34690/479

PREPRINTS

Mar. 2026 Kornev M.I.,
Buchstaber, Ochanine, Krichever, and Witten genera
arXiv:2603.21118

Oct. 2025 Buchstaber V.M., Kornev M.I.
Algebraic n-Valued Monoids on $\mathbb{C}P^1$, Discriminants and Projective Duality
arXiv:2510.14010

Aug. 2025 Kornev M.I.
On the Classification of n-Valued Monoids and Groups of Order 3
arXiv:2508.04454

May 2025 Buchstaber V.M., Kornev M.I.
n-Valued Groups, Kronecker Sums, and Wendt's Matrices
arXiv:2505.04296

Jan 2024 Kornev M.I.
n-valued Coset Groups and Dynamics
arXiv:2401.12718

CONFERENCE TALKS

Aug. 2025 *n-Valued Groups, Kronecker Sums, and Wendt Matrices*
joint work with Victor Buchstaber
5th Conference of Mathematical Centers of Russia

May 2024 *Homology Spheres, Acyclic Groups and Kan-Thurston Theorem*
Conference "From Analysis to Homotopy Theory" , University of Greifswald, Greifswald, Germany

Apr. 2023 *Dynamics and multivalued groups*
Conference "SIMC Youth Race", Steklov Mathematical Institute of RAS, Moscow, Russia

Oct. 2022 *Inconcretness of homotopy categories*
International School "Toric topology, combinatorics and data analysis", Euler International Mathematical Institute, Saint Petersburg, Russia

Apr. 2022 *Acyclic groups and spaces*
29th International Scientific Conference for Undergraduate and Graduate Students and Young Scientists "Lomonosov", Moscow, Russia

Nov. 2021 *Acyclic groups and Kan-Thurston theorem*, International School "Toric Topology and Combinatoric", Sirius Mathematics Center, Sochi, Russia

SEMINAR TALKS

May 2026	*Buchstaber genus and formal groups* Seminar "Toric Topology, Combinatorics, and Homotopy Theory" jointly with the Laboratory of Topology and Complex Networks of the Faculty of Mathematics, HSE University, Moscow Center for Continuous Mathematical Education (MCCME), Moscow, Russia
Dec. 2025	*Fermat hypersurfaces, projective duality, and algebraic n-valued monoids* Seminar of the Department of Geometry and Topology "Geometry, Topology and Mathematical Physics" (Novikov Seminar), Steklov Mathematical Institute of RAS, Moscow, Russia
Dec. 2025	*Discrete and Algebraic n-Valued Groups* Seminar on Discrete Geometry and Geometry of Numbers, Moscow State University, Moscow, Russia
Nov. 2025	*n-Valued Algebraic Monoids, Cubics, and Discriminants* The Seminar of International Laboratory of Algebraic Topology and Its Applications, HSE University, Moscow, Russia
May 2025	*n-Valued Groups, Kronecker Sums, and Wendt Matrices* Joint work with Victor Buchstaber Seminar of the Department of Geometry and Topology "Geometry, Topology and Mathematical Physics", Steklov Mathematical Institute of RAS (Novikov Seminar), Moscow, Russia
Apr. 2023	*Acyclic groups and Kan-Thurston theorem* Seminar by I. G. Lysenok and A. L. Talambutsa "Seminar on geometric group theory", Steklov Mathematical Institute of RAS, Moscow, Russia
Dec. 2022	*Fundamental groups of homology spheres* Seminar "Geometry, Topology and their applications" at Sobolev Institute of Mathematics, Online

GRANTS AND AWARDS

Prizes, fellowships and grants

2021 – 2023	*The Russian Science Foundation Grant No. 23-11-00143*
2019	*Winner of the olympiad on Partial Differential Equations* Moscow State University
2018	*Winner of the olympiad on Ordinary Differential Equations* Moscow State University

STUDY SEMINAR TALKS

Texas Tech Quantum Homotopy Seminar

Dec. 2024	*An Introduction to Cartan Geometries*
Mar. 2024	*Euler-Lagrange dynamics via the ∞-jet bundle* (2 talks)

Seminar On Homotopy Theory and Applications at Independent University of Moscow

Dec. 2023	*Principal ∞-bundles*

Apr. – May 2023	*A mini-course on the ∞-category of motivic coarse spaces* (4 talks)
Apr. 2023	*Elements of unstable motivic homotopy theory*
Feb. – Apr. 2023	*A mini-course on ∞-cosmoi* (4 talks)
Dec. 2022	*Applications of Reedy theory*
Nov. 2022	*Weighted (co-)limits, Gambino's theorem, Reedy model structures*
Nov. 2022	*Homotopically final functors and examples of calculations with homotopy (co-)limits*
Nov. 2022	*Nerve—realization adjunction and bar construction*
Apr. 2022	*A series of 4 talks on sheaves of sets*
Apr. 2022	*Kan extensions. Additional topics*
Oct. 2021	*The categorical meaning of the Kan-Thurston theorem*
Jul. – Aug. 2021	*Quillen's theorem, a closed model structure on sSet*

Study And Scientific Seminar "Geometry And Topology" at Moscow State University

Sep. 2020	*Andreev's theorem on hyperbolic polyhedra*
Nov. 2019	*The intersection form of 4-manifolds*
May 2019	*Nilmanifolds and Massey product in bigraded cohomology rings*

CONFERENCES, WORKSHOPS AND SCHOOLS

May 2024	*Conference "From Analysis to Homotopy Theory"* University of Greifswald, Greifswald, Germany
Jul. 2023	*Summer School "Algebra and Geometry"* HSE University, Steklov Mathematical Institute of RAS, Suzdal, Russia
Jul. 2023	*Summer school on non-commutative and derived geometry of matrix factorisations* Centre of Pure Mathematics, Moscow Institute of Physics and Technology, Dolgoprudny, Russia
Jun. 2023	*Conference on algebra, algebraic geometry, and number theory on the occasion of academician Igor Rostislavovich Shafarevich 100th birthday* Steklov Mathematical Institute of RAS, Moscow, Russia
Apr. 2023	*Conference "Geometry, topology and mathematical physics" on the occasion of the 85th anniversary of S. P. Novikov and the 80th anniversary of V. M. Buchstaber* Steklov Mathematical Institute of RAS, Moscow, Russia
Mar. 2023	*Conference "SIMC Youth Race"* Steklov Mathematical Institute of RAS, Moscow
Oct. 2022	*International School "Toric topology, combinatorics and data analysis"* Euler International Mathematical Institute, Saint Petersburg, Russia
Jul. 2022	*Summer School "Algebra and Geometry"* HSE University, Steklov Mathematical Institute of RAS, Suzdal, Russia
Apr. 2022	*29th International Scientific Conference for Undergraduate and Graduate Students and Young Scientists "Lomonosov"* Lomonosov Moscow State University, Moscow, Russia
Nov. 2021	*International School "Toric Topology and Combinatorics"* HSE University, Sirius Mathematics Center, Sochi, Russia

Nov. 2020	*Topology and Geometry of Group Actions* HSE University, online
Nov. 2019	7th Summer School On Geometrical Methods Of Mathematical Physiques Landau Institute for Theoretical Physics, Lomonosov Moscow State University, Chernogolovka, Russia
Nov. 2019	*Geometry, Topology And Mathematical Physics* Landau Institute for Theoretical Physics, Lomonosov Moscow State University, Chernogolovka, Russia
Nov. 2018	*Geometry, Topology And Mathematical Physics* Landau Institute for Theoretical Physics, Lomonosov Moscow State University, Chernogolovka, Russia
Jul. 2018	*Summer school "Modern Mathematics"* HSE University, Lomonosov Moscow State University, Independent University of Moscow, Dubna, Russia
Jul. 2017	*Summer school "Modern Mathematics"* HSE University, Lomonosov Moscow State University, Independent University of Moscow, Dubna, Russia

TEACHING

May 2022 – 2024	*Teaching various courses (e. g., Algebraic Geometry, Algebraic Topology, Abstract Algebra, Differential Geometry, Linear Algebra, Probability Theory, Discrete Mathematics, Calculus, etc.) on Profi* `https://profi.ru/profile/KornevMI3/`
Sep. 2016 – 2022	*Little Mech-Math courses* An olympiad Math class at Lomonosov Moscow State University `http://mmmf.msu.ru/about/person.html`

SERVICE

June 2021 – 2024	*Co-organization of the study seminar "Homotopy Theory and Applications" at Independent University of Moscow* `https://researchseminars.org/seminar/HomotopyTheoryAndApps`

REFERENCES

Victor Buchstaber,
Steklov Mathematical Institute of RAS
`buchstab@mi-ras.ru`

LANGUAGES

Russian (native)

English (fluent)

French (intermediate)

Latin (basic)